Thoughts on Mathematical Values

I think we sometimes overlook how much freedom we still have to decide what kinds of mathematics we want to pursue, what we find valuable, and what mathematical traditions we want to preserve. The development of AI may change what is possible, but it does not have to determine what we choose to value.

I used to have a relatively neutral or optimistic attitude towards AI in mathematics. If AI could produce genuinely good mathematics, I saw no reason not to appreciate it, provided that mathematicians were willing to understand the results. I also thought that conceptual understanding and mathematical judgment might remain particularly difficult for AI, and that the mathematical community might naturally shift its values towards aspects of mathematics in which AI was less capable.

I still find these considerations reasonable. But I no longer think our mathematical values should depend on predictions about what AI will or will not be capable of.

What Makes Mathematics Valuable?

There are many different mathematical theories, methods, and research directions, all of which can be good mathematics. But mathematicians do not value them equally. Different fields have their own traditions, tastes, and standards of what counts an interesting contribution. These judgments are shaped not only by mathematical difficulty or usefulness, but also by history, individual personalities, and intellectual traditions.

Mathematics is not simply a collection of correct statements. It is a human intellectual activity in which people develop ideas, build theories, and seek understanding. The mathematical tastes and insights of particular mathematicians have shaped entire schools of mathematics, each with its own ideas about what is interesting or worth pursuing.

When it comes to AI, the mathematical community has the freedom to decide what kinds of contributions it chooses to recognize. In particular, I think an independent human proof should still deserve the recognition traditionally associated with a first proof, even if an AI-generated proof already exists.

But whether being first should be valued so highly is itself worth questioning. In some mathematical communities, considerable importance is attached to being the first to solve a problem, and AI may create new competitive pressures. But the point of mathematics is not simply to compete over who solves a problem first. Different methods and ways of understanding the same result can be equally important whether or not they are the first or not. Sometimes it is precisely a different proof that opens the way to important developments.

Of course, there is a reasonable attitude that good mathematics should be appreciated regardless of how it was produced. I still sympathize with this view. But the mathematical community has considerable freedom to decide what counts as good mathematics and what kinds of contributions it values. This distinction becomes especially important now since valuing “good” mathematics is no longer necessarily the same as valuing human mathematical activity or understanding.

Mathematics and Artistic Independence

Artists and musicians often have strong ideas about what makes their work valuable. They may refuse to follow commercial trends and insist on pursuing their artistic ideals despite the uncertainty this implies. Mathematics, like art and music, is not only about producing results. The experience, individual judgment, and the intellectual traditions passed between generations are themselves extremely valuable. There is something worth defending in the freedom to pursue mathematics according to one's judgement of what is valuable. It is equally important to preserve a mathematical culture in which such choices are respected and appreciated.

Maintaining intellectual independence can be difficult, especially for younger mathematicians facing career uncertainty and competitive pressures. Some worry that AI will make academic competition even more unfair, but I am not convinced that the evaluation system is necessarily so vulnerable. Uncertainty existed long before AI. Academic careers have never depended entirely on mathematical ability or the quality of one's work. Research fashions, connections, advisors' recommendations, and luck have always mattered. Moreover, academic hiring considers more than publications, it often includes qualities that cannot simply be replaced by AI-generated results. AI does not necessarily make the situation worse.

Of course, we should not ignore new concerns and difficulties caused by AI. Senior mathematicians and academic institutions have considerable influence over which contributions are recognized and rewarded. They can help shape a mathematical culture in which mathematicians do not feel compelled to adopt particular research practices merely to remain competitive. More importantly, they can help preserve and develop mathematical traditions that value human understanding and human intellectual effort.

Different mathematical communities may also respond differently to AI. Certain fields that emphasize concrete results, technical improvements, or priority may face stronger competitive pressures.

I am not sure how much these differences come from the differences in mathematics itself and how much from the cultures of individual fields. Perhaps AI will amplify differences in mathematical values that already exist between different fields.

There is no universal solution to these difficulties. But perhaps part of intellectual independence is accepting that doing what we value does not always bring the recognition or outcomes we hope for.

Developing Mathematical Understanding

AI can still be very useful in many ways, but I think we should be careful about what we use AI for. I do not think we should ever adopt a research style in which humans provide only conceptual judgment while leaving the mathematical details to AI. At the very least, authors should be able to check and understand all the details themselves. But understanding is itself difficult to judge. Knowing how much we really understand is a skill that develops with experience. Sometimes it takes being repeatedly challenged or reminded that we have not understood something to develop such judgment.

I remember fondly how my commutative algebra mentor could often look at a complicated proof and say that there was a much simpler argument once one understood what was really going on. I also learned from working with different professors that there are different ways of engaging with details to develop understanding. The details of a proof can also reveal a lot about how a mathematician understands mathematics.I also feel lucky to have inherited a distinctive mathematical perspective from my algebra mentor, which has given me directions to explore for years to come. I suppose such insights come from deep understanding and also have something to do with individual mathematical personalities. They develop through sustained engagement with mathematical details, including work that may not appear directly useful. Much of our mathematical judgment develops through engaging with seemingly trivial details.

This is why I strongly oppose relying on AI-generated mathematics without understanding. Part of this comes from my belief in the value of mathematical understanding, but also from strong attachment to a particular mathematical tradition I highly value.

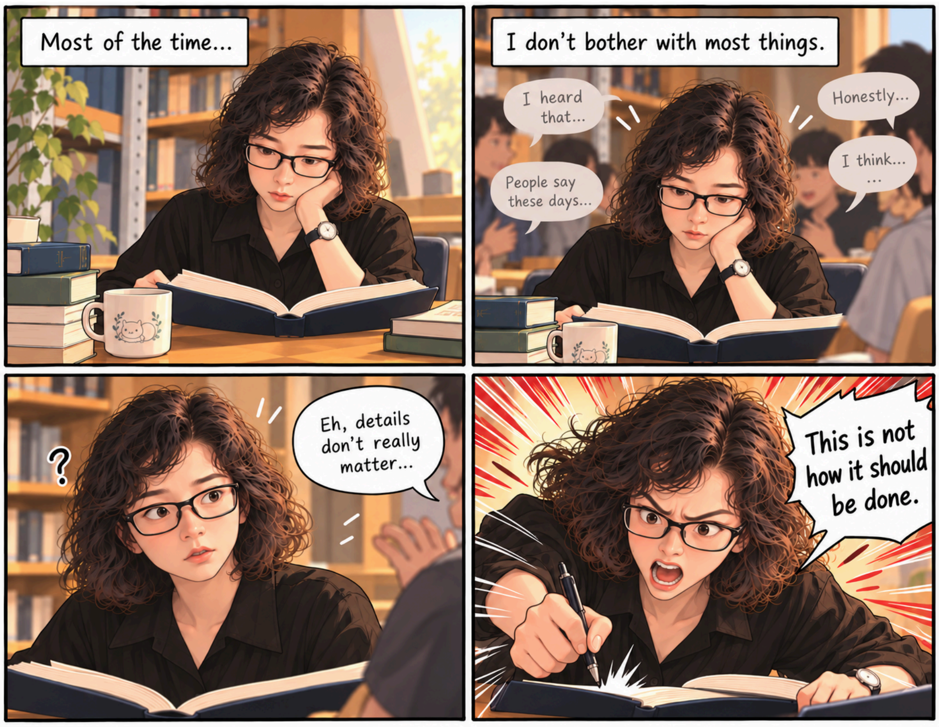
At the same time, I generally trust mathematicians who have developed mature mathematical judgment to decide how best to use these tools. I am not suggesting that most mathematicians use AI without understanding, and I recognize that many ways of using AI in research are perfectly respectable.

Continuing to Enjoy Mathematics

Perhaps the most important thing is that mathematics should remain something we genuinely enjoy doing.

Mathematical values need not remain unchanged forever. But changes in what we value should come from our understanding of mathematics and our judgement about what we value, not simply from what technology becomes capable of producing.

Perhaps mathematicians should be a little more stubborn about the things we find valuable. The point is not simply to produce as much mathematics as possible, or to compete with whatever becomes capable of producing it. It is also to continue finding mathematics worth thinking about, understanding, and enjoying.


Most of the time...
I don't bother with most things.
I heard that...
Honestly...
People say these days...
I think... ...
?
Eh, details don't really matter...
This is not how it should be done.

L.X.